

Words to equations



1 For each of the following word statements:

- i Write the word statement as an equation, where p is the unknown number.
- ii Solve the equation.

- a The sum of an unknown number and 39 is 83.
- b 75 is equal to the sum of an unknown number and 42.
- c The product of 3 and an unknown number is 18.
- d -7 multiplied by an unknown number is 84.
- e 17 less than an unknown number is 25.
- f The quotient of an unknown number and 8 is -12 .
- g 2 times the sum of an unknown number and 5 is 14.
- h The sum of a number and 2 divided by 3 is 6.

2 A bag of lollies contains 80 lollies. The lollies are shared evenly between 10 children.

Let n represent the number of lollies each child receives.

- a Write down the equation that represents the relationship between the total number of lollies and the number of lollies each child receives.
- b Hence, solve the equation.

3 Justin divides a deck of cards into 7 even piles. There are 12 cards in each pile.

Let c represent the total number of cards in the deck.

- a Write down the equation that represents the relationship between the total number of cards in the deck and the number of cards in each pile.
- b Hence, solve the equation.

4 Tricia gives Tom 25 marbles from her collection. Tricia now has only 44 marbles in her collection.

Let m represent the number of marbles she had before giving some away.

- a Write down the equation that represents the relationship between the initial number of marbles and the number of marbles left in her collection.
- b Hence, solve the equation.

- 5 Lisa bought a hat and a dress for a total of . Let h represent the cost of the hat and d represent the cost of the dress.

- a Write down the equation that represents the relationship between the total cost and the individual costs of the hat and dress.
- b Hence, find the cost of the hat if the cost of the dress is \$56.

Tables

- 6 Consider the word statement " y is equal to the product of 7 and x ".

- a Write an equation in the form $y = \dots$ that describes the word statement.

x	1	2	3	4	5
y					

- b Hence, complete the table.

- 7 Consider the word statement " v is equal to the product of -8 and u ".

- a Write an equation in the form $v = \dots$ that describes the word statement.

u	1	2	3	4	5
v					

- b Hence, complete the table.

- 8 Consider the word statement " q is equal to p less than 12".

- a Write an equation in the form $q = \dots$ that describes the word statement.

p	-2	-1	0	1	2
q					

- b Hence, complete the table.

- 9 Consider the word statement "19 subtracted from r is equal to s ".

- a Write an equation in the form $s = \dots$ that describes the word statement.

r	-2	-1	0	1	2
s					

- b Hence, complete the table.

- 10 Consider the word statement " y is equal to the product of 3 and the sum of x and 4".

- a Write an equation in the form $y = \dots$ that describes the word statement.

x	1	2	3	5	10
y					

- b Hence, complete the table.

11 Consider the word statement " v is equal to the product of 5 and the sum of u and 2".

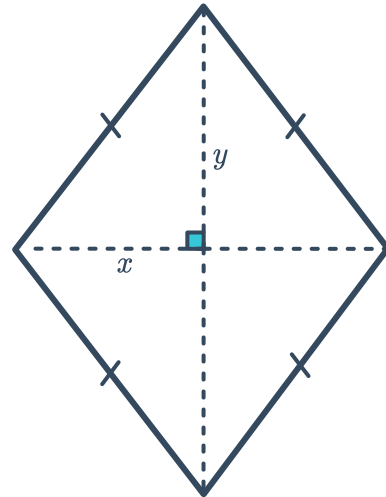
- a Write an equation in the form $v = \dots$ that describes the word statement.
- b Hence, complete the table.

u	1	2	3	5	10
v					

Formulas

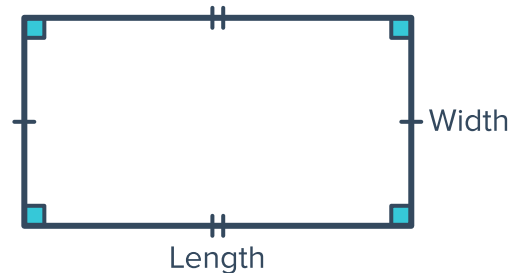
12 The area of a rhombus is given by the formula $A = \frac{1}{2}xy$, where x and y are the lengths of the diagonals:

- a Find the area of a rhombus which has short and long diagonal lengths of 4 cm and 6 cm respectively.
- b A particular rhombus has a short diagonal length $x = 6$ m and area $A = 33 \text{ m}^2$. Find the value of y .



13 The area of a rectangle is given by the formula $A = \text{length} \times \text{width}$:

- a If the length of a rectangle is 9 m and its width is 3 m, find its area.
- b The area of a particular rectangle is 45 m^2 . If its length is 9 m, determine the width of the rectangle.

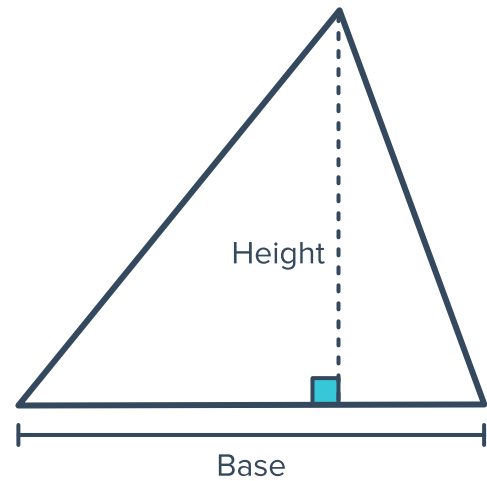


14 The area of a triangle is given by the

formula $A = \frac{1}{2} \times \text{base} \times \text{height}$:

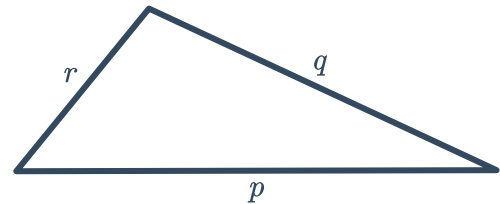


- a If the base of a triangle is 8 mm and its height is 6 mm, find its area.
- b A triangle with a height of 4 mm has an area of 16 mm². Find the length of the base of the triangle.



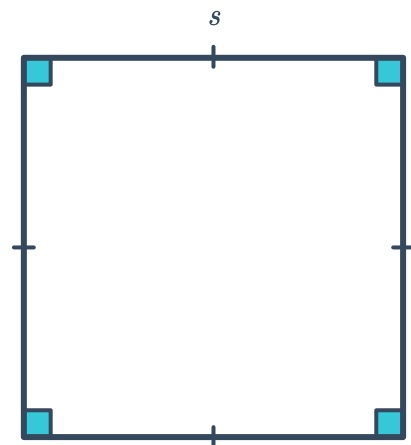
15 The perimeter of a triangle with sides of lengths p , q and r is given by the formula $P = p + q + r$:

- a Find P if the length of each of its three sides are $p = 9$ mm, $q = 7$ mm and $r = 5$ mm.
- b If $P = 40$ mm and the length of each of the two sides are $p = 10$ mm and $q = 12$ mm, what is the length of the third side r ?

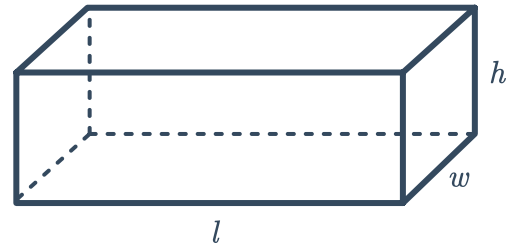


16 The perimeter of a square with side lengths of s is given by the formula $P = 4 \times s$:

- a Find P if the length of each side is 3 m.
- b Find s if the perimeter is $P = 40$ m

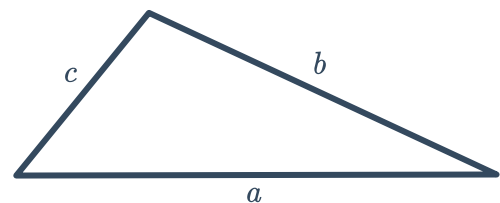


- 17 The volume of a rectangular prism is given by the formula $V = l \times w \times h$, where l , w and h are the dimensions of the prism:



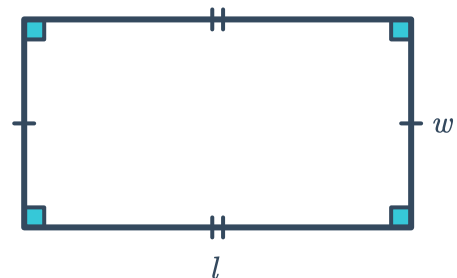
- a Given that a rectangular prism has a length of 9 m, a width of 4 m and a height of 7 m, find its volume.
- b The volume of a particular rectangular prism is 140 cm^3 . If it has a length of 10 cm and a width of 7 cm, find the height, h .

- 18 The perimeter of a triangle can be calculated with the formula $P = a + b + c$, where a , b and c are the three side lengths of the triangle. If a triangle has a perimeter of $P = 37 \text{ cm}$ and side lengths $a = 14 \text{ cm}$ and $b = 13 \text{ cm}$, find the length of the side c .



- 19 The perimeter of a rectangle is given by the formula $P = 2 \times (l + w)$, where l is the length and w is the width:

- a Find the perimeter of the rectangle which has a length of $l = 6 \text{ cm}$ and width of $w = 5 \text{ cm}$.
- b If a certain rectangle has length $l = 7 \text{ cm}$ and perimeter $P = 24 \text{ cm}$, find the width w .



- 20 The speed of a plane can be calculated using the formula $S = \frac{D}{T}$, where D is the distance travelled, T is the time taken and S is the speed.

- a If a plane travels 3600 kilometres in 6 hours, find its speed.
- b If the speed of the plane is 700 km/hr , find the distance travelled after 4 hours.

- 21 The formula to convert temperature from Celsius to Fahrenheit is $F = 32 + \frac{9C}{5}$.

- a If $C = 35$, find the value of F .
- b If $F = 212$, find the value of C .